

9.3.1.5. Scrieti cele 2 forme canonice simple si verificati:

x	y	z	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

$$M_0 = x \vee y \vee z$$

$$m_1 = \bar{x} \bar{y} z$$

$$m_2 = \bar{x} y \bar{z}$$

$$M_3 = x \vee \bar{y} \vee \bar{z}$$

$$M_4 = \bar{x} \vee y \vee z$$

$$m_5 = x \bar{y} z$$

$$m_6 = x y \bar{z}$$

$$M_7 = \bar{x} \vee \bar{y} \vee \bar{z}$$

$$FCC(f) = M_0 \wedge M_3 \wedge M_4 \wedge M_7 =$$

$$= (x \vee y \vee z) \wedge (x \vee \bar{y} \vee \bar{z}) \wedge (\bar{x} \vee y \vee z) \wedge (\bar{x} \vee \bar{y} \vee \bar{z})$$

$$FCD(f) = m_1 \vee m_2 \vee m_5 \vee m_6 =$$

$$= \bar{x} \bar{y} z \vee \bar{x} y \bar{z} \vee x \bar{y} z \vee x y \bar{z}$$

	x		$\bar{x}$	
y		m_6	m_2	
	$\bar{y}$	m_5		m_1
	z	$\bar{z}$	z	

$$max_1 = y \bar{z}$$

$$max_2 = \bar{y} z$$

$$M_f = \{max_1, max_2\}$$

$$C_f = \{max_1, max_2\}$$

$$C_f = M_f \Rightarrow \text{caz 1}$$

9.3.2.5. Simplificati următoarele fct. booleene de 4 variabile date prin formule dis-junctive, utilizând diagrama Veitch'

$$f(x_1, x_2, x_3, x_4) = x_1 x_2 x_3 x_4 \vee x_1 x_2 \bar{x}_3 x_4 \vee x_1 x_2 x_3 \bar{x}_4 \vee \bar{x}_1 x_2 \bar{x}_3 \bar{x}_4 \vee \bar{x}_1 x_2 x_3 \bar{x}_4 \vee x_1 \bar{x}_2 x_3 \bar{x}_4 \vee x_1 \bar{x}_2 \bar{x}_3 \bar{x}_4 \vee x_1 \bar{x}_2 x_3 x_4$$

	x_1	$\bar{x}_1$	
x_2	m_{15}	m_{13}	x_4
	m_{14}		m_4
			m_6
$\bar{x}_2$	m_{10}	m_8	$\bar{x}_4$
		m_9	x_4
	x_3	$\bar{x}_3$	x_3

$$max_1 = m_4 \vee m_6$$

$$max_2 = m_{10} \vee m_8$$

$$max_3 = m_{15} \vee m_{13}$$

$$max_4 = m_{15} \vee m_{14}$$

$$max_5 = m_8 \vee m_9$$

$$max_6 = m_6 \vee m_{14}$$

$$max_7 = m_{10} \vee m_{14}$$

$$max_8 = m_{13} \vee m_9$$

$$M_f = \{max_1, max_2, \dots, max_8\}$$

$$C_f = \{max_1\}$$

$$M_f \neq C_f \neq \emptyset \Rightarrow \text{caz 2}$$

$$g(x_1, x_2, x_3, x_4) = max_1$$

$$h_1(x_1, x_2, x_3, x_4) = max_3 \vee max_7 \vee max_5$$

$$h_2(x_1, x_2, x_3, x_4) = max_8 \vee max_2 \vee max_4$$

$$f'_1(x_1, x_2, x_3, x_4) = g(x_1, x_2, x_3, x_4) \vee h_1(x_1, x_2, x_3, x_4) = max_1 \vee max_3 \vee max_7 \vee max_5$$

$$f'_2(x_1, x_2, x_3, x_4) = g(x_1, x_2, x_3, x_4) \vee h_2(x_1, x_2, x_3, x_4) = max_1 \vee max_8 \vee max_2 \vee max_4$$

9.3.4.5. Quine-McCluskey

$$f(x_1, x_2, x_3) = x_3(\bar{x}_1 \vee x_2) \vee x_1 \bar{x}_3$$

Prin metoda tabeli $\rightarrow f(x_1, x_2, x_3) =$

0
1
0
0
1
1
1
1

Grupul $x_1 x_2 x_3$

I	1 1 1	m_7
II	1 0 1	m_5
	1 1 0	m_6
III	0 0 1	m_1
	1 0 0	m_4
$\text{IV} = \text{I} + \text{II}$	1 - 1	$m_7 \vee m_5$
	1 1 -	$m_7 \vee m_6$
$\text{V} = \text{II} + \text{III}$	1 - 0	$m_6 \vee m_4$
	- 0 1	$m_5 \vee m_1$
	1 0 -	$m_5 \vee m_4$